Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/28/2005
PI Name:	Fedorov, Andrei G Ph.D.		
Project Title:	MEMS-ENABLED PROCESSING OF LIQUID FUELS FOR DISTRIBUTED POWER GENERATION USING FUEL CELLS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30332	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Degertekin, Levent (Georgia Institute of Technology)		
Grant/Contract No.:	NNC04GA12G		
Performance Goal No.:			
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Task Description:	Fuel cell based portable power sources offer opportunities for high efficiency and ultra-high density distributed power generation, which is essential for enhanced capabilities and survivability of the future long duration space missions as well as for extra vehicular activities (EVA) of astronauts. A key element of the fuel cell power sources is the fuel processor, which should have a capability to produce hydrogen from a variety of hydrocarbon fuels stored in a liquid phase for highest power density. To this end, this project focuses on the novel portable MEMS hydrogen generator with improved functionality and low cost achieved through innovative system integration approach and by exploiting advantages pertinent to fluid mechanics, heat and mass transfer, and catalysis on micro/nano scale. The proposed microreactor has multifuel processing capability achieved by using unique piezoelectrically-driven fuel atomization technology, it operates in autothermal regime that does not require external heating, it exhibits high fuel conversion and selectivity to hydrogen, and it features low “skin” temperature which is essential from thermal management prospective. To establish the fundamental bases for optimal design and operation of the proposed catalytic microreactor, a ground-based three-year experimental study of the microreactor and its key components is undertaken. The key objectives include (1) design, fabrication, and testing of the autothermal MEMS catalytic reactor integrated with hydrogen separating membrane, (2) design, fabrication, and experimental characterization of the novel micromachined fuel ejector array for energy-efficient atomization and mixing of liquid fuels, and (3) integration and experimental characterization of an integrated microreformer that combines a microreactor, a hydrogen separating membrane, and an atomizer/droplet ejection array. The key objectives include (1) design, fabrication, and testing of the autothermal MEMS catalytic reactor integrated with hydrogen separating membrane, (2) design, fabrication, and experimental characterization of the novel micromachined fuel ejector array for energy-efficient atomization and mixing of liquid fuels, and (3) integration and experimental characterization of an integrated microreformer that combines a microreactor, a hydrogen separating membrane, and an atomizer/droplet ejection array. Space exploration is an energy intensive enterprise that demands power from Watt-scale for sensors and electronics to MegaWatt-scale for spaceship service and propulsion. Development of highly efficient and environmentally friendly power sources is equally important for the Earth-based applications. In addition, volume and mass limitations on spacecraft make the power density requirements very stringent (approaching 1.0-1.5 kWh per kilogram of fuel), especially for portable electronic and communication devices that are being used by astronauts during the extra vehicular activities. These requirements put an ever-increasing demand on development of small-scale, highly efficient “palm” power sources. The existing battery technologies become major obstacle to advances in performance of portable energy-intensive devices (primarily because of the limited lifetime between the recharges), and the fuel cell-based energy sources appear to be the only viable candidate for battery replacement. Fuel cell-based portable power sources offer opportunities for high efficiency and ultra-high density distributed power generation, which is essential for aerospace and terrestrial applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Pollution reduction and development of environmentally friendly energy conversion technology
Task Progress:	Fuel cells offer opportunities for high efficiency and ultra-high density distributed power generation, which is essential for many aerospace and terrestrial applications. A key element of the fuel cell power sources is the fuel processor, which should have a capability to produce hydrogen from a variety of hydrocarbon fuels. To this end, we are developing a novel MEMS hydrogen generator with improved functionality achieved through innovative system integration approach by exploiting advantages of transport and catalysis on micro/nano scale. The microreactor has multifuel processing capability achieved by using unique ultrasonic fuel atomization technology; it operates in autothermal regime without external heating with high fuel conversion and selectivity to hydrogen; and it utilizes in-situ separation of hydrogen from the reaction products via reactor-integrated sub-micrometer thick Palladium-Silver membrane. The key project objectives (sub-tasks) include (1) design, fabrication, and testing of the autothermal MEMS catalytic reactor and hydrogen separating membrane, (2) design, fabrication, and characterization of the micromachined fuel ejector for atomization of liquid fuels, and (3) integration and experimental characterization of an integrated microreformer that combines a microreactor, hydrogen separating membrane, and an atomizer. During the reporting period, we have achieved a number of critical advances in all three key aspects of the project. In particular, the first prototype of the MEMS microreactor has been designed and fabricated for the experimental performance characterization of the reverse-flow operation. In parallel, a theoretical analysis of the reactor operability and optimal operation has been performed that revealed the unique advantages of the reverse-flow autothermal microreactors as compared to the unidirectional counterparts. Of those, an unlimited reactor throughput while maintaining an autothermal (i.e., self-sustained without external heating) operation and its robustness (i.e., little dependence of the hydrogen yield on small variation in the feed composition) are the two main advantages that provide strong impetus for pursuing the reverse-flow microreactor concept. We have also perfected the fabrication process for ultra-thin (nanoscale) Pd-Ag membranes suitable for highly selective separation of hydrogen from the reaction mixture. This resulted in repeatable manufacturing of membranes which meet stringent requirements of structural and chemical stability to be suitable for experimental testing. To support the experimental work, a theoretical model of hydrogen permeation has been developed. The results of the analysis indicates that desorption of hydrogen on the low pressure side of the membrane may become the rate limiting step, rather than solid-phase atomic hydrogen diffusion as commonly thought for the conventional (macroscale) membranes. Finally, we have continued our work on understanding the fluid physics and performance optimization of the ultrasonic MEMS ejectors needed for fuel atomization prior to introduction to the reactor. The scaling law governing transition from drop-on-demand to continuous jet operation of the ejectors has been discovered through combination of high resolution visualization and theoretical analysis. Furthermore, a novel approach for improving power transfer efficiency for running the ejectors at low power input has been developed and experimentally demonstrated.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Articles in Peer-reviewed Journals	Kaisare NS, Lee JH, Fedorov AG. "Hydrogen generation in a reverse-flow microreactor I: Model formulation and scaling." AIChE Journal 2005 Aug;51(8):2254-64. http://dx.doi.org/10.1002/aic.10492 , Aug-2005
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Presentation	Fedorov, A. G. "Transport and Chemical Processing on Small Scales" Milliken and Company, Spartanburg, South Carolina, USA Jun-2005
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